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The Wealthfront Generation

It was a late October morning in 2014 and the Palo Alto offices of automated investment service Wealthfront were bustling with activity. As the executive team prepared to discuss the launch of a new, more expansive “Direct Indexing Platform,” CEO Adam Nash (HBS 2001) took a moment to reflect on the dizzying pace of accomplishments the company had achieved since he joined less than two years before. The firm’s assets under management (AUM) had grown from less than \$100 million in January 2013 to over \$1.5 billion, placing the company among the top-100 independent registered investment advisors (RIAs)^a in the United States. Innovations like daily tax-loss harvesting were gaining wide acceptance, and the company had just announced a financing round of approximately \$64 million, bringing total funds raised to over \$129 million, with a post-money valuation of over \$700 million.

Wealthfront created a fully automated, low-cost, online investment platform designed to target the over 90 million Americans often referred to as the “millennial generation.” The company’s user-friendly platform offered investors the ability to invest in a diversified portfolio of index funds based on their personal risk tolerance. The service continuously monitored those investments and implemented best practices like rebalancing, dividend reinvestment, and tax-loss harvesting. Founders Andy Rachleff and Dan Carroll positioned Wealthfront as a disruptive force within the investment management industry. By dramatically lowering the minimum investment required, they sought to democratize access to sophisticated investing.

In less than three years since its launch in December 2011, Wealthfront had achieved substantial growth, yet Nash believed the company had only scratched the surface of an enormous opportunity. He and the cofounders were resolute about several aspects of the business: putting the client first, focusing on millennials, fully automating the service, delighting clients through product innovation, and staying independent as they scaled. Despite the latest financing, the company still had not spent the \$35 million it raised six months earlier and was in a strong financial position to pursue hyper-growth. Nash, a veteran of Silicon Valley giants such as Apple, eBay, and LinkedIn, felt Wealthfront also could be a world-class, profitable, and valuable business. Yet Nash faced a variety of strategic

^a An RIA was defined in the 1940 Investment Advisers Act as a “person or firm that, for compensation, is engaged in the act of providing advice, making recommendations, issuing reports or furnishing analyses on securities, either directly or through publications.” An RIA had a fiduciary duty to his or her clients, which meant that he or she had a fundamental obligation to provide suitable investment advice and always act in the clients’ best interests. Source: www.investopedia.com, accessed December 15, 2015.

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questions. Most importantly, he needed to decide whether Wealthfront should maintain its focus on the consumer channel or expand into business-to-business channels.

Everyone Deserves Sophisticated Financial Advice

The concept for Wealthfront grew out of Rachleff's and Carroll's independently held, shared belief that there was an injustice in the everyday investors' lack of access to sophisticated investment management. Such high-quality advice often came with high minimum account size requirements and expensive fees. Rachleff, a seasoned Silicon Valley-based venture capitalist and Stanford Graduate School of Business lecturer, had the opportunity to work with sophisticated investment managers both as a direct investor and as a trustee of the University of Pennsylvania, where he served as Vice Chairman of its Endowment Investment Committee. He recalled, "It bothered me that most people don't have access to outstanding investment managers. I realized I could do a social good by solving that problem." Thousands of miles away in Chicago, Carroll, a former bond trader and financial data provider employee, grew frustrated watching his parents' investment accounts managed poorly, with high fees. He reflected, "I realized that what happened to my parents was probably just a small example of what was going on everywhere. I wanted to find a way to help people like my parents manage their investments."

Before meeting, each created his own service to provide high-quality, affordable investment management to all investors. Based on the recommendation of a former student familiar with Carroll's efforts, Rachleff approached Carroll to join forces. Carroll recalled, "I initially ignored Andy's calls because I didn't know much about venture capital. But I am glad I finally engaged." The two came together in 2008 to hatch kaChing, an online marketplace for high-performing U.S. equity investment managers.¹ Their work on kaChing bolstered their belief in the power of software and automation to enable their vision. Conversations with people who didn't sign up for kaChing provided insights into investor preferences. Rachleff recalled, "We consistently heard, 'I'd rather you manage my *entire* portfolio adequately and inexpensively than manage only a portion of my portfolio superbly.' That was a light-bulb moment."

With lessons from kaChing and private funding in hand, Rachleff and Carroll launched Wealthfront in December 2011. Carroll recalled, "We had the classic chicken-and-egg problem. Investors view AUM as a proxy for quality. How do you attract clients if you don't have any assets?" An early Wealthfront investor, retired Kleiner Perkins Caufield & Byers partner Doug Mackenzie, suggested the company target young technology workers. "Go after the millennials," he advised. He believed this demographic would care more about the quality of the user experience and the company's team than its AUM.

The \$7 Trillion Opportunity

The founders seized an opportunity created by two demographic shifts in the U.S.: the retirement of the baby boomer generation (those born between 1946 and 1964), and the coming into adulthood of the millennial generation (those born between 1980 and 2000). According to U.S. Census statistics, baby boomers made up 23.7% of the U.S. population, while millennials accounted for 28.7%.² The Brookings Institute estimated that one in three adults would be a millennial in the year 2020.

Nash believed that long-established brokerage houses such as Merrill Lynch and Charles Schwab, as well as independent RIAs, focused almost exclusively on the baby-boomer generation, a group that controlled \$16 trillion in assets. He estimated the average age of a Charles Schwab customer was

54, and that of a Merrill Lynch customer, 63. The average RIA was approximately 50.³ Nash felt that the existing industry's high minimum account requirements forced a sales-driven process that targeted older investors with assets of at least \$1 million. Traditional investment advisory firms charged an average annual advisory fee of 1% or more on managed assets. Some alleged that firm profits were enhanced by steering clients into investment products that paid financial incentives. Rachleff noted, "Baby boomers have been raised with the expectation that high-quality service isn't cheap. They are usually more than happy to pay a premium to gain access to someone who can hold their hand through difficult decisions." Nash elaborated:

You could almost say that the incumbents have a \$16 trillion *problem*. They employ scores of human advisors, and these advisors only have bandwidth for 50–100 clients each. The advisor needs to earn a living, and the firm needs to earn a profit on top of that. You end up with a certain revenue per client number that you need to meet as a firm to make this work. So while these firms may recognize millennials are a good bet for the future, they cannot afford to go after them, given their cost structure. Every dollar they spend going after a millennial is a dollar not spent servicing the baby boomer.

Millennials were the first generation to have had access to the Internet during their youth. They were tech-savvy, with 85% of U.S. millennials owning smartphones and three-quarters having accounts on social media sites.^{4,5} In terms of their financial lives, millennials were three times more likely to hold their assets in cash than in stock, and 39% of millennials selected cash as their preferred investment vehicle, the highest percentage of any generation.⁶ Media reports often speculated that their apparent risk aversion was due to living through the Great Recession, coupled with a natural conservatism. Nash disagreed, "I'm skeptical of reports that millennials are afraid to invest. My grandmother lived through the Great Depression. The number of millennials walking around with \$500 phones doesn't speak Depression-era mentality to me. After two market crashes, they are very cynical about the sales-driven process of the traditional industry." A 2014 Blackrock study indicated that millennials were more confident about their financial future than any other age group. The same study showed that millennials spent up to seven hours per month reviewing their stock portfolios compared to the national average of four hours per month.⁷

In Nash's view, the millennial generation had been overlooked by incumbent investment advisory firms, and yet this untapped generation presented a significant opportunity. He noted that millennials controlled an estimated \$1 trillion–\$2 trillion in assets, a figure he projected would increase to \$7 trillion by 2020 through income generation and, to a lesser degree, inheritance. Carroll noted, "Millennials, especially those employed in tech, are often paid in equity grants and in bonuses. They get a lot of cash windfalls they don't know what to do with."

Wealthfront divided the investing public into two groups: "delegators," who preferred to involve an intermediary, often an individual or firm, to achieve their investment goals, and "do-it-yourselfers," who enjoyed actively managing their own assets. They believed the split was three-quarters delegators and one-quarter do-it-yourselfers, with millennials leaning toward delegating. Nash opined, "Millennials have a lot of passions—career, family, hobbies, and travel. There's a long list of what they want to spend time on, and managing their money is not one of them." The Wealthfront website was designed to appeal to delegators. Rachleff elaborated:

The assumption to date has been that if you put a financial product or information on the Web, it must be targeted at do-it-yourselfers. But we're seeing something different with millennials. For those who want to delegate, handing that delegation over to

software is a more trustworthy experience than handing it over to a financial advisor, because software is rational, scalable, and transparent. It's a generational shift. If you're over 50, you want to delegate to a person; if you are under 35, you distrust the sales person. Millennials like the idea that software is working for them 24/7.

Wealthfront's focus on millennials drew criticism from some industry observers who speculated the company would lose millennial clients once those clients aged and faced major life events such as marriage, divorce, birth of children, death of a spouse, and retirement. Rachleff quipped, "I don't think that generation is going to turn 40 and all of a sudden want to talk to somebody on the phone about their money. They're not going to switch from rap to symphony when they age. It's a culture thing."

Designing an Industry-Defining Company⁸

Wealthfront was designed from the outset to be different from existing Wall Street firms. That started with its business model. In order to attract millennials who worked in the technology field, the company implemented a classic Internet "freemium" business model^b inspired by cloud storage company Dropbox. Just as Dropbox initially offered its users 2 gigabytes (GB) managed for free and 250 additional megabytes (MB) for every referred customer, Wealthfront initially managed a client's first \$25,000 for free and offered an additional \$5,000 managed for free for every referred new client. As was the case with Dropbox, the referred client also received an additional \$5,000 in funds managed for free if the client took advantage of an electronic referral. In contrast, traditional financial service firms typically charged small accounts a higher fee, which declined as assets were added to the account.

The Wealthfront platform offered millennials (with a minimum account size of \$5,000) a significant value proposition, that of investing their money at a reasonable fee (0.25% per year on the amount in excess of assets managed for free), with best-practice investment tools, leveraged by automation. "Transparency drives trust with millennials. We do exactly what we say we'll do," Nash noted. The user experience was simple, engaging, and offered an interface via desktop or mobile device. Rachleff said, "Millennials are used to subscribing to all their services electronically. To them, investing is just another service that should be automated in the cloud."⁹ Unlike traditional investment firms, Wealthfront did not employ sales professionals or open retail locations.

The Power of Viral Acquisition

The company's go-to-market strategy started with investment seminars delivered to Internet companies poised to go public. Wealthfront's seminars were purely educational and appealed to engineers who were wary of Wall Street. "No sales pitches like the investment bank seminars," noted Nash. Credibility was built by courting the recommendation of "investment influencers," employees at these companies recognized by their peers as knowledgeable sources of information about investing. The aforementioned invitation system accelerated the word of mouth generated by the seminars. In addition, the company leveraged content marketing through high-quality blog posts containing data-driven and actionable personal finance advice and investment white papers posted on its website, all of which were amplified by a meaningful presence on social media.

^b The term was used to describe a business model in which a company offered basic services *free* of charge and *premium* services for a fee. The phrase became associated with software firms that gave basic functionality for free trial and charged for additional capabilities. Source: <http://www.investopedia.com/terms/f/freemium.asp>, accessed December 15, 2015.

“Millennials love to share the services they enjoy,” said Rachleff. Market research firm Social Chorus noted that 95% of millennials said friends were the most credible source of product information.¹⁰ As of late 2014, one-eighth of individuals who received electronic referrals opened an account. Nash reflected, “Having worked on viral systems in the past, I found this amazing. Generally, individuals don’t feel comfortable talking about money because they view it as awkward and too personal. They feel culpable if they make a referral and something goes wrong. And yet, even unoptimized, Wealthfront had an exceptional conversion rate. No investment service had proven viral in financial services before. When most of your customer acquisition is free, it’s a good sign you have product-market fit.”

Simple, Transparent, and Fair Pricing

Wealthfront’s advisory fee was approximately one-quarter of that charged by traditional financial advisors. Wealthfront recognized that a software-based investment solution could be offered at a low marginal cost, and affordability was part of its mission. The company also felt it aggressively undercut the more than 50 basis point cost structure it assumed for incumbent financial advisory firms.

Investors also paid the embedded fee on the exchange traded funds (ETFs) employed in their portfolios (on average 11 basis points). Nash reflected, “People often assume that young people are exclusively focused on price, but there is more than that. There is a strong trend towards pricing that is perceived as simple, transparent, and fair. No gimmicks like a monthly fee. No gotchas. And no special discounts for the wealthy.”

Wealthfront in Fall 2014

By the fall of 2014, Wealthfront had a strong executive team, a world-class chief investment officer, a well-defined corporate culture, and clients that expressed a high degree of satisfaction with its service. The company was funded by several prominent venture capital firms as well as several dozen well-known Silicon Valley individual investors. (See **Exhibit 1** for list.)

Team

Wealthfront had 42 Palo Alto-based employees, 24 of whom were software engineers. According to Nash, thanks to its mission-driven culture, employee turnover was extremely low relative to other startups in the area. The board of directors included Nash, Rachleff (Executive Chairman), Carroll (Chief Strategy Officer), former Netflix Chief Financial Officer Barry McCarthy, and Index Ventures Partner Mike Volpi.

Nash had been an early client and advocate of the service, opening his first account in January 2012. Nash was an executive-in-residence at Menlo Park, California-based venture firm Greylock Partners, when he first met Rachleff. Nash recalled:

I’d been looking at a lot of fintech investments. I was frustrated with many of the companies I saw. They got the business model wrong *and* they didn’t have an effective distribution strategy. Much of my background was in consumer software where success has to do with getting distribution right, and turning it into a product that can’t easily be replicated by the next startup. I thought Wealthfront was the right business model. I knew if we could get it to scale, we’d have a great company. I’m also a big believer in

consumer brands. It's great to have that direct link to your customer. It removes all conflict of interest and allows you to build a better business over time.

In January 2013, Nash joined as Chief Operating Officer, later becoming CEO in January 2014 after the rapid growth of the company in 2013. Initial CEO Rachleff became the Executive Chairman of the Wealthfront board. Nash recruited some of Silicon Valley's best consumer Internet and software engineering talent, many recognized for playing important roles growing companies such as Facebook, LinkedIn, and Twitter. Carroll described: "In building the team, we took a 'let's get the band back together' approach, as often happens in Silicon Valley." Nash elaborated: "I didn't just want good. I wanted *the best*. We needed to marry world-class financial talent with the best engineers and designers in Silicon Valley."

The investment team included industry luminaries, PhDs, and chartered financial analysts (CFAs). Burton Malkiel, noted economist and emeritus professor at Princeton University, joined Wealthfront as Chief Investment Officer in late 2012. In his 1973 best-selling book, *A Random Walk Down Wall Street*, Malkiel proposed passive investing as the most effective long-term investment strategy. Assuming it was impossible to outperform the market, he argued that investors should try to mimic broad market indexes and focus their attention on the three activities that could have an impact on their performance: portfolio diversification, lowering fees, and minimizing taxes. As an advocate of index investing, Malkiel served as a member of the board of directors at Vanguard for 28 years. Malkiel said, "Software-based advice is the only way the little guy can get personalized professional financial advice without paying an arm and a leg for it. I've been working in investor education all my life. In Wealthfront, I've found people who believe as I do, and can make a high-quality service possible for everyone."¹¹

The investment advisory board comprised many well-known financial experts, including Charley Ellis, founder of Greenwich Associates; Paul Pleiderer, Stanford economist and risk management specialist; Meir Statman, Santa Clara professor and behavioral finance expert; and Larry Cohen, CEO of Seven Bridges Advisors, a financial advisory service that served family offices. The research team, led by Avery Moon, Vice President of Research and Engineering, included several PhDs who worked closely with the advisory board and created the algorithms and methodologies on which Wealthfront based its software. (See **Exhibit 2** for team biographies.)

Culture

The Wealthfront office culture was informal and open, similar in feel to a software company rather than a traditional financial services firm. Open debate on key decisions was encouraged, and all team members were invited to express their ideas. The office's open floor plan and the communal kitchen were conducive to such exchange. The firm was client-centric, with the statement, "We believe our clients are our one true north," appearing first on the list of company values emblazoned on a centrally located 14-foot wall. (See **Exhibit 3** for the Wealthfront values board.)

Wealthfront was purposefully built as a mission-driven organization. Nash reflected, "Our people could get jobs wherever they want—Dropbox, Facebook, Google, Uber—but they choose to work here because of the people and the mission." Rachleff mused, "We think of ourselves as a software company that delivers an investment service rather than an investment company that builds software. Given the extremely tight market for top software engineering talent in the Bay Area, we see a recruiting advantage in offering employees an opportunity to potentially impact the financial well-being of millions of people." Moon elaborated:

Our engineers are not just building widgets that need to meet quality control. They are craftspeople creating software that codifies the culture of trust that permeates our organization. I believe the best talent wants to get involved in regulated industries where there hasn't been a lot of disruption, like healthcare, education, and finance. That gets them excited about doing something that matters. There's a saying that culture eats strategy for breakfast. It's true. Our engineers want to be at a company that values innovation and doesn't treat software as a commodity.

Nash spent his early time as CEO identifying and analyzing key business metrics. At an early all-staff meeting, he shared, "There are two simple things we need to do: acquire clients and delight them, because if we delight them, they will never leave us, keep saving money with us, and tell their friends." He developed a simple dashboard with four metrics and placed it in a central location. Nash reflected, "In the software industry, you're used to dealing with very smart people. What they need from leadership is just two things. They need to know what game we're playing and how to keep score."¹² This clarity of purpose and transparency enhanced internal communications and influenced transparency in external communications; the Wealthfront website provided investors significant detail regarding portfolios and methodologies.

Nash's four metrics were visitor traffic, conversion rate, funding rate, and net deposits per client. From January 2013 to September 2014, visitor traffic had grown by over 170% and conversion had improved by 78%, all while maintaining a high funding rate and increasing net deposits from clients. As a result, AUM had grown from \$97 million in January 2013 to over \$1.5 billion in October 2014.

Statistics

As of October 2014, Wealthfront served approximately 14,000 clients. Client account sizes ranged from \$5,000 (the stated account minimum) to over \$10 million, with an average account size of \$90,000 (up from \$40,000 in 2013). The average initial account size was \$30,000. On average, accounts grew to \$100,000 by their one-year anniversary. Several small endowments and scholarship funds opened accounts with Wealthfront.org, which managed up to \$1 million in nonprofit money free of charge. New accounts were most commonly funded with cash or a concentrated stock position earned from working at a company that recently went public. (See **Exhibit 4** for chart of AUM from 2011 to 2014.)

Customer acquisition costs had been minimal, as free organic and viral acquisition channels attracted more than 80% percent of all new clients. AUM grew in three ways. First, new investors signed up for the service. Second, existing investors added assets to their accounts (on average, Wealthfront clients with \$100,000 or more AUM added 20% of their original deposit amount to their portfolio on a quarterly basis). Third, portfolio values grew from positive market performance.

Clients ranged in age from 18 to 93 years old. Sixty percent of clients were under 35, 30% were between 36 and 50, and 10% were baby boomers or older. Given its viral distribution, Wealthfront quickly grew to have clients in all 50 U.S. states, and three military postal codes. Like the general millennial population, Wealthfront clients tended to be concentrated in urban centers. (See **Exhibit 5** for map of concentration.) On average, Wealthfront managed approximately 15% of its clients' disclosed investible assets.

Client feedback was overwhelmingly positive. Nash noted, "Our clients appreciate that their accounts are customized and straightforward. They tell us they're happy because we do exactly what we said we'll do." Carroll noted, "We've clearly nailed the trust question. The other day we had someone open a \$3.7 million account without placing one phone call." Wealthfront executives

believed trust was enhanced by virtue of the company's status as an SEC-registered investment advisor. Assets were held in accounts in the clients' name at a third-party brokerage firm where the accounts were insured up to \$500,000 by the Securities Investor Protection Corporation (SIPC). Regulators and independent auditors ensured that investor assets were not comingled, and cash and securities held in investor accounts were not loaned out or borrowed by Wealthfront or its brokerage partner.

Investment Platform

Wealthfront's investment platform was built to maximize the long-term net-of-fee, after-tax, real investment return for each client's particular tolerance for risk. The company's investment methodology was rooted in modern portfolio theory (MPT). Moon noted, "We're standing on the shoulders of giants as we build upon a well-defined set of financial principles."

Asset class diversification Wealthfront believed that a long-term view of one's investments, governed by steadfast adherence to asset-allocation models, would result in superior long-term outcomes compared to those based on emotion or market timing. According to leading financial services market research firm DALBAR, the average equity fund investor performed 4.32% worse than the S&P 500 from 1992–2011 by trying to time the market. Typically, these underperforming investors sold shares after a market decrease and bought shares after the market rose.¹³

While many traditional wealth managers directed investor portfolios to three asset classes—U.S. stock, foreign stock, and U.S. government bonds¹⁴—Wealthfront considered 11 asset classes for use in its client portfolios. They included equities (U.S. stock, foreign developed stock, emerging market stock, and dividend growth stock), fixed-income (U.S. government bonds, corporate bonds, emerging market bonds, and municipal bonds), and "real" assets (assets that provided protection in an inflationary environment, such as natural resources, real estate, and Treasury Inflation-Protected Securities, or TIPS).^c Investor accounts were invested across a diversified basket of these asset classes. Fully invested Wealthfront accounts did not hold cash as part of the asset allocation. Wealthfront believed that adding more asset classes than traditional wealth managers typically raised the expected return on both taxable and retirement accounts by as much as 0.6% per annum for the same amount or risk.

To maximize return for every level of risk (as measured by variance), Wealthfront used the mean-variance portfolio optimization (MVO) framework to determine the optimal mix of asset classes within a portfolio. The first input to MVO was the estimate of the expected return for each asset class. The research team estimated expected returns based on the capital asset pricing model (CAPM)^d and then used the Black-Litterman model^e to adjust CAPM long-term return expectations according to its views. The second set of inputs to MVO was the estimated standard deviation of returns for each asset class. To estimate each asset class's standard deviation (volatility), the team considered its long-term historical standard deviation, its short-term standard deviation, and the expected volatility implied by its pricing in the options markets. To estimate the third set of inputs, correlation, they

^c TIPS were Treasury bonds whose coupons and principal payments grew with inflation as measured by the Consumer Price Index. Therefore, their payouts were indexed to inflation.

^d CAPM was an equilibrium model of the relationship between risk and return that characterized a security's expected return based on its beta with the market portfolio. Source: Jonathan Berk and Peter DeMarzo, *Corporate Finance*, 2nd ed. (2011), p. G-3.

^e The framework was developed by Fischer Black and Robert Litterman in 1992 while working at Goldman Sachs. The model derived expected return parameters from equilibrium returns and manager views. Source: Wealthfront white paper.

considered long-term and short-term historical correlations. (See **Exhibits 6a** and **6b** for asset class weights by risk level.)

Wealthfront assigned minimum and maximum percentage allocations for each asset class in order to gain appropriate diversification. Portfolios were long only. The tax treatment of each asset class was taken into consideration when constructing portfolios. For example, taxable accounts did not contain real estate, corporate bonds, and emerging market bonds because they were tax inefficient, while retirement accounts did not contain U.S. government bonds, municipal bonds, and natural resources because their inclusion did not increase return expectations for the same amount of risk.

Passive investing Wealthfront believed in low-cost, long-term investing through passive, indexed investing. The goal of passive investing was to meet the overall (or benchmark) return on each asset class, not to beat it. The goal of active investing, on the other hand, was to outperform the benchmark return on each asset class. Passive investing was achieved through investing in index mutual funds or ETFs, while active investing was achieved through purchasing individual securities or actively managed mutual funds and investment vehicles such as hedge funds. Managers of actively managed investment funds typically charged a fee of 0.6% to 2.0% of AUM and, in the case of hedge funds, an additional share (typically in the 20% range) of any gains in excess of the return on a passive benchmark, compared to fees of 0.04% percent to 0.2% for funds employing passive strategies.

Passive investing strategies were growing in popularity. In the first nine months of 2014, investor outflows from actively managed funds totaled \$80 billion, compared to \$60 billion inflows into passively managed funds.¹⁵ Research showed that, after accounting for fees, actively managed mutual funds had underperformed the passively managed Vanguard S&P 500 ETF (VTI) by an average of 2.1% per year, before taxes, from 1980 to 2000.¹⁶

Wealthfront accessed each asset class through ETFs. ETFs were launched in the 1990s as investment vehicles similar to open-end index mutual funds that were traded on an exchange like a traditional stock. ETFs owned the same underlying securities and in the same proportion as the underlying index. Ownership of the underlying securities was fractional, much like mutual funds. However, unlike mutual funds that traded only once a day at the end of a trading day at the net asset value (NAV) of the fund at market closing,^f ETFs traded throughout the trading day. Other benefits of ETFs included transparency and lower costs.^g ETFs had grown in popularity as an investment vehicle during their 15 years in existence. With nearly \$2 trillion in AUM, ETFs accounted for approximately 15% of the investment fund industry in the U.S.¹⁷ (See **Exhibit 7** for a list of ETFs and their associated fees on the Wealthfront platform, and **Exhibits 8a** and **8b** for sample model portfolios by investor risk tolerance level.)

Dividend-based rebalancing A portfolio's asset allocation could become unbalanced due to gains or losses among asset classes. Wealthfront monitored its clients' allocations on a continuous basis to ensure they were in line with the recommended portfolios. Wealthfront's rebalancing was triggered when an account met a certain threshold of variance from the optimum portfolio (rather than the time-based approach traditional managers often applied, sometimes in an ad hoc fashion).

^f The NAV of a fund or ETF is defined as the total value of its assets minus the value of its liabilities—such as fees owed to the management company. For equity mutual funds, asset value is typically calculated based on the closing market price of the stocks held in the fund.

^g For more information on the history of ETFs, see Luis Viceira and Alison Wagonfeld, "Barclays Global Investors and Exchange Traded Funds," HBS No. 208-033 (Boston: Harvard Business School Publishing, November 8, 2007).

Rebalancing could be tax inefficient because it typically implied that an investor who reaped gains in one asset class would sell a portion of that asset class to maintain the desired portfolio balance, thus incurring taxes. In contrast, the Wealthfront service considered the tax impact when rebalancing portfolios. For example, any inflows into a client's account were used to purchase an underweighted asset class (rather than being invested pro rata into the recommended asset classes automatically). Similarly, dividends were reinvested in the underweighted asset class rather than the security that generated them to help balance portfolios. The service also sold the tax lots with the least gains to minimize taxes upon withdrawal of funds.

Daily tax-loss harvesting U.S. individual investors were subject to taxes on realized capital gains and dividends in their taxable accounts, which could have a significant impact on portfolio returns. Investors could reduce the impact of taxes on their returns by engaging in a practice known as "tax-loss harvesting" in which ETFs that declined in value were sold to realize a loss that could be used to lower taxes. The sold ETFs were replaced with ETFs that tracked highly correlated indexes so as to maintain the risk and return characteristics of the portfolio. Investors could use these realized capital losses to offset up to \$3,000 of ordinary income each year and an unlimited amount of capital gains. Losses remaining after matching against gains and income could be carried over indefinitely for future years. Tax savings were higher for residents of high income tax states. Nash recalled:

When we researched this issue, we found a lot of ad hoc approaches. Brokers or individuals might sell losers at year end, but that offered a small fraction of the benefit of looking for losses to harvest daily. Coding the process in software enabled us to offer daily tax-loss harvesting to a broad audience at no additional cost.

In September 2012, Wealthfront became the first automated investment service to offer daily tax-loss harvesting across ETFs. In order to qualify for the service, clients needed to open taxable accounts with at least \$100,000. When first launched, Wealthfront estimated the incremental annual benefit of tax-loss harvesting could be 0.93%. (See **Exhibits 9a** and **9b** for an excerpt from a white paper for explanation of the methodology and calculation of tax alpha.)

In December 2013, Wealthfront became the first automated investment service to offer tax-loss harvesting *within* an equity index. Clients with over \$500,000 had access to the service, called "Direct Indexing," which replaced the ETF that would have been used for the U.S. stock slice of a client's portfolio (Vanguard VTI) with the 500 stocks that comprised the S&P 500 plus an ETF that represented small cap U.S. stocks. Direct Indexing allowed for automated tax-loss harvesting at the individual stock level, which could add an incremental annual after-tax benefit of 0.70% over what could be achieved with daily tax-loss harvesting only.

Single stock diversification In April 2014, Wealthfront introduced the single stock diversification service (SSDS), which targeted employees of companies that had recently gone public. The service allowed investors with a significant portion of their personal wealth concentrated in a single stock to gradually sell their position over time using the daily volume weighted average price (VWAP), a sophisticated sales strategy typically only available to institutions.

Innovation at the "Press of a Button"

Elliott Shmukler, Vice President of Product and Growth, positioned Wealthfront as a service, rather than a product: "When you buy a product, that product is inert; it's static. Whereas a service evolves over time. Think of an age-dated retirement fund also known as a 'target date fund.' Do you have a relationship with that fund? No. People have relationships with services. The beauty of software is we have control over the entire client experience. We solve a problem once and push it

out to all accounts automatically.” Nash recognized that while Wealthfront was not the first entrant to the new field of automated investment services, it was the firm’s innovation culture that differentiated it from others: “Wealthfront has a track record of rolling out the innovations that define the category. In this business, having features like tax-loss harvesting first has significant economic value for clients.”¹⁸ (See **Exhibit 10** for Wealthfront’s innovation timeline.)

Moon noted that his engineering team’s continuous deployment capability (the ability to push code to production many times each day rather than the industry standard of once weekly) was applauded throughout the startup ecosystem. This capability facilitated rapid client adoption of enhanced feature offerings. When Wealthfront increased the number of fixed-income asset classes available on the platform from one to five in 2013, 60% of clients chose to upgrade to the expanded offering within 48 hours after e-mail notification. Nash commented, “You might expect to see that kind of adoption in a software setting. But not in the world of investment management where you’d first need to train the sales force on what the change is and how to explain it to clients.” Similarly, in the first three months after launch, investors placed over \$250 million into Direct Indexing accounts.

Scaling Wealthfront

As the meeting was about to start, Nash gazed at the large screen that displayed the hockey-stick-shaped graph of AUM growth over time. He was proud of what the team had accomplished and excited to build the future. At the same time, he faced a long list of near-term operational priorities, each of which competed for his attention and the firm’s resources. He wanted to continue hiring a strong team, further product innovation, increase the rate of client acquisition, and build out the infrastructure. While the firm’s financial position was strong, each of these initiatives required managerial oversight and coordination. “We needed to get the different areas moving in parallel,” Nash said. Carroll reflected, “When you start to scale the business, you have to learn to walk and chew gum at the same time. It sounds easy, but it’s a different rhythm. In the beginning, you are laser-focused on doing one or two things well, but the challenges become more complex as you grow the team and take on multiple initiatives at the same time.”

Nash had divided the company into four development programs, each with its own metrics and priorities. It focused on growth, client experience, investment products, and infrastructure. He often contemplated what proportion of the company’s resources should be allocated to each program.

Expanding the Reach to More Investors

Wealthfront’s focus on the millennial generation had proven relatively contrarian to the approach taken by its competitors. Most new entrants and incumbents targeted clients of every demographic. Nash was convinced that Wealthfront’s strategic advantage lay in the “innovator’s dilemma” facing the incumbents, as their economic focus had to be on the \$16 trillion held by baby boomers. The average baby-boomer client account size was twice that of a millennial client. Nash wondered if Wealthfront should seek faster asset growth by pursuing older investors.

Extending the Business Model

In 2014, the consensus among industry insiders was growing that a “hybrid” human advisor/automated investment service model was the best approach to managing clients’ investments. In other words, it would make more sense for companies like Wealthfront to deliver tools to existing advisors rather than try to directly acquire clients. Most of Wealthfront’s competitors had announced private-label platforms for RIAs. The RIA market was large and growing, with over \$2 trillion in

AUM.¹⁹ Offering a Wealthfront-powered “white label” solution to these RIAs might greatly expand the investors on the Wealthfront platform and further the mission of democratizing access to sophisticated investment management. If Wealthfront did not pursue the RIA channel, it could miss out on a significant source of new assets. Nash asked himself if it was worth diluting the company’s focus and direct-to-consumer model to expand into this new channel.

An Evolving Competitive Landscape

While Nash believed that Wealthfront offered compelling differentiation from its peers, he was also mindful of recent developments in the competitive landscape. He believed that New York-based Betterment, a close competitor, was growing rapidly by mimicking features, like the tax-loss harvesting that had driven Wealthfront’s popularity. A number of new financial technology (fintech) startups were also targeting the millennial demographic, and incumbents were waking up to the threat posed by Wealthfront and others. The day before Wealthfront announced its latest financing, Charles Schwab Corporation announced its own automated investment service, Intelligent Portfolios, which would be available to investors free of advisory fees. Schwab’s new platform would reportedly offer a variety of automated features similar to what Wealthfront had offered in 2012. Schwab also announced plans to introduce a white-label version for RIAs shortly thereafter. RIAs already represented more than \$1.2 trillion in Schwab assets. While Nash was skeptical that Schwab’s offering would truly put the customer’s interests first as Wealthfront did, he could not ignore the signal of this development. (See **Exhibit 11** for an overview of the competitive landscape.)

Looking Ahead

Nash was optimistic: “When I look into the future, I see Wealthfront as a membership experience. Clients will feel a part of a company that is changing the industry for the better, bringing lower cost and higher value to all investors.” He continued, “I believe that in 10 years, Wealthfront will be an incredibly valuable brand. When people look back, they’ll say, ‘Of course. It was obvious.’ Just as the largest generation of Americans the country had ever seen began reaching the age where they start investing, the financial crisis dramatically damaged all of the existing financial service brands. With ample, long-term venture capital available, of course you’d build a new service, designed from the ground up to be a software company, to meet the needs of this new generation in the 21st century.”

Exhibit 1 List of Investors (venture firms and individuals)

Venture Capital Firms			
Date	Lead Investor(s)	Round	Amount
October 2014	Spark Capital	Series D	\$64 million
April 2014	Index Ventures Ribbit Capital	Series C	\$35 million
March 2013	Greylock Partners Index Ventures	Series B	\$20 million
December 2009	DAG Ventures	Series A	\$7.5 million
December 2008	Various Individuals	Angel	\$3 million
Private Individuals and Professional Affiliation			
Adam D'Angelo, Quora	Dave Beirne, Fantx	Ken Goldman, Yahoo!	Owen Tripp, Grand Rounds Health
Adam Fischer, Howard Fisher Associates	Dave Morin, Path	Kevin Rose, Google Ventures (former)	Paul Kedrosky, SK Ventures
Ali Pincus, One Kings Lane	David Hahn, LinkedIn	Louis Eisenberg, Facebook	Peter Pham, Science, Inc.
Andy Dunn, Bonobos	Doug Mackenzie, Radar Partners	Marc Andreessen, Andreessen Horowitz	Reed Hundt, FCC Chairman (former)
Angela Zaeh, Facebook	Gil Petchina, Wikia	Marissa Mayer, Yahoo!	Ronnie Lott, SF 49'ers (former)
Ankur Pansari, Artillery	Harris Barton, H. Barton Asset Management	Mark Pincus, Zynga	Ryan Roslansky, LinkedIn
Barry McCarthy, Spotify	Hunter Walk, Google (former)	Matt Mullenweg, Automatic	Satya Patel, Twitter (former)
Ben Horowitz, Andreessen Horowitz	Javier Olivan, Facebook	Matt Papaikos, Facebook	Scott Roberts, LinkedIn
Bruce Dunlevie, Benchmark	Jeff Jordan, Andreesen Horowitz	Matt Wyndowe, Uber	Sharmila Mulligan, ClearStory Data
Chris Morace, Jive Software	Jeff Weiner, LinkedIn	Mike Jones, Science, Inc.	Tim Ferriss, Author
Cipora Herman, S.F. 49'ers	Kay Luo, LinkedIn	Mike Schroepfer, Facebook	Tim Kendall, Pinterest
Dan Shapero, LinkedIn		Mike Vernal, Facebook	

Source: Company website, <https://www.wealthfront.com/investors>, accessed May 2016.

Note: Includes only those who were or had been invested as of October 2014.

Exhibit 2 Management Team Biographies

Adam Nash, President and CEO. Adam is a proven advocate for development of products that go beyond utility to delight customers. Adam joined Wealthfront from Greylock Partners, where he was an Executive-in-Residence. Prior to Greylock, he was VP of Product Management at LinkedIn, where he built the teams responsible for core product, user experience, platform, and mobile. Adam has held a number of leadership roles at eBay, including Director of eBay Express, as well as strategic and technical roles at Atlas Venture Preview Systems and Apple. Adam holds an MBA from Harvard Business School as well as BS and MS degrees in Computer Science from Stanford University.

Andy Rachleff, Executive Chairman. Andy serves as a member of the board of trustees and vice chairman of the endowment investment committee for University of Pennsylvania and as a member of the faculty at Stanford Graduate School of Business, where he teaches courses on technology entrepreneurship. Prior to Wealthfront, Andy co-founded and was general partner of Benchmark Capital, where he was responsible for investing in a number of successful companies including Equinix, Juniper Networks, and Opsware. He also spent ten years as a general partner with Merrill, Pickard, Anderson & Eyre (MPAE). Andy earned his BS from University of Pennsylvania and his MBA from Stanford Graduate School of Business. He has frequently been named to Forbes magazine's The Midas List as a top tech venture investor.

Dr. Burton G. Malkiel, Chief Investment Officer. The Chemical Bank Chairman's Professor of Economics, Emeritus, and Senior Economist at Princeton University, joined Wealthfront in November 2012. Dr. Malkiel is the author of the widely read investment book, *A Random Walk Down Wall Street*, which helped launch the low-cost investing revolution by encouraging institutional and individual investors to use index funds. Dr. Malkiel, also the author of *The Elements of Investing*, is one of the country's leading investor advocates. Before moving to Princeton, where he was also chairman of the Economics Department, Dr. Malkiel was dean of the Yale School of Management and William S. Beinecke Professor of Management Studies from 1981 until 1988. He is a past president of the American Finance Association and the International Atlantic Economic Association. He is also a past appointee to the President's Council of Economic Advisors. In addition, he serves on the boards of directors and advisory boards of several corporations and is a past board member of Vanguard Group.

Dan Carroll, Founder and Chief Strategy Officer. Dan founded Wealthfront to bring client-centric, transparent and low cost financial advice to the retail investor. During the financial crisis of 2008, he was determined to fix his family's precarious position with their financial advisor, and this experience ultimately led to the mission driven model that Wealthfront offers today. Dan was an investor at a young age and began his career working in fixed income at financial research firm FactSet. He holds a BS in finance from University of Arizona and was named one of the "Best young Tech Entrepreneurs under 30" by *Bloomberg Businessweek* in 2001.

Rick Foreman, CFO and VP Business Operations. Rick joined Wealthfront in 2009 from Bloom Energy, where he was VP of Finance & Administration for 4 years. Prior to Bloom, he was an Executive-in-Residence with Benchmark Capital, and for six years was VP of C-Cube Microsystems, a leader in digital video. Rick began his Silicon Valley career by joining Cypress Semiconductor three months after its inception, and for ten years held senior management positions in operations, finance, and information technology. Rick holds a BS in Mechanical Engineering from Villanova University, an MS in Systems Engineering from the University of Pennsylvania, and an MBA from Wharton Graduate School.

David Fortunato, CTO. Before joining Wealthfront in 2009, David was a software engineer at Ning, working on a variety of components from front end features to back end performance. At Wealthfront, David has worked extensively on deployment, monitoring, and trading systems. David holds a BA in Computer Science and Economics from Amherst College.

Avery Moon, VP of Research and Engineering. Avery thrives leading world-class teams in hyper-growth. Avery joined Wealthfront from LinkedIn, most recently as Sr. Director of Engineering, where

he built and led the engineering teams responsible for Growth and Core features. Avery also led key infrastructure projects, ranging from multiple data center architecture to distributed development and integration. Prior to LinkedIn, he held leadership and technical roles at EMC and two venture-backed startups. He graduated summa cum laude from Eller College of Management at University of Arizona, and holds a BA in Economics and a BS in Applied and Industrial Mathematics from University of Arizona.

Elliot Shmukler, VP of Product and Growth. Elliot is an accomplished Silicon Valley “growth hacker,” having led the strategy and execution behind LinkedIn’s growth from 20 million to more than 200 million members. He joined Wealthfront from LinkedIn, where, as Sr. Director of Product Management, he also oversaw much of the consumer product experience, including Profile, Search, Endorsements, and People You May Know. Previous to LinkedIn, Elliot held technical and leadership roles at eBay, About.com, Sombasa Media, and Microsoft. Elliot holds an MBA from Stanford Graduate School of Business, where he was an Arjay Miller Scholar, and holds a BA magna cum laude in Computer Science from Harvard University.

Brian Dennen, Chief Compliance Officer and CEO of Wealthfront Brokerage Corporation. Brian oversees all client investment activities. Brian provides nearly 15 years of compliance and trading experience, most recently serving as Managing Director & Chief Compliance Officer of Livevol Securities. Brian holds a BS in Management Science from UC San Diego and FINRA series 4, 7, 24, 27, 63, 65, and 99 licenses.

Ali Rosenthal, VP Strategic Partnerships. Ali has a proven track record and passion for building world-class companies and services through partnership and business development. Ali joined Wealthfront from MessageMe, where she served as COO. Previously, she was an Executive-in-Residence with Greylock Partners after serving almost six years at Facebook. Ali joined Facebook in early 2006 when it had less than 5 million users and 100 employees and drove business development efforts including Facebook’s relationships with over 350 mobile operators and OEMs. During her tenure, Facebook Mobile’s user base grew from under 100,000 to over 250 million users. Ali also previously worked at Goldman Sachs and General Atlantic Partners. She serves on the boards of AutoNation and KQED. Alison holds an MBA from Stanford and a BA from Brown University.

Source: Company website, <https://www.wealthfront.com/investors>, accessed May 2016.

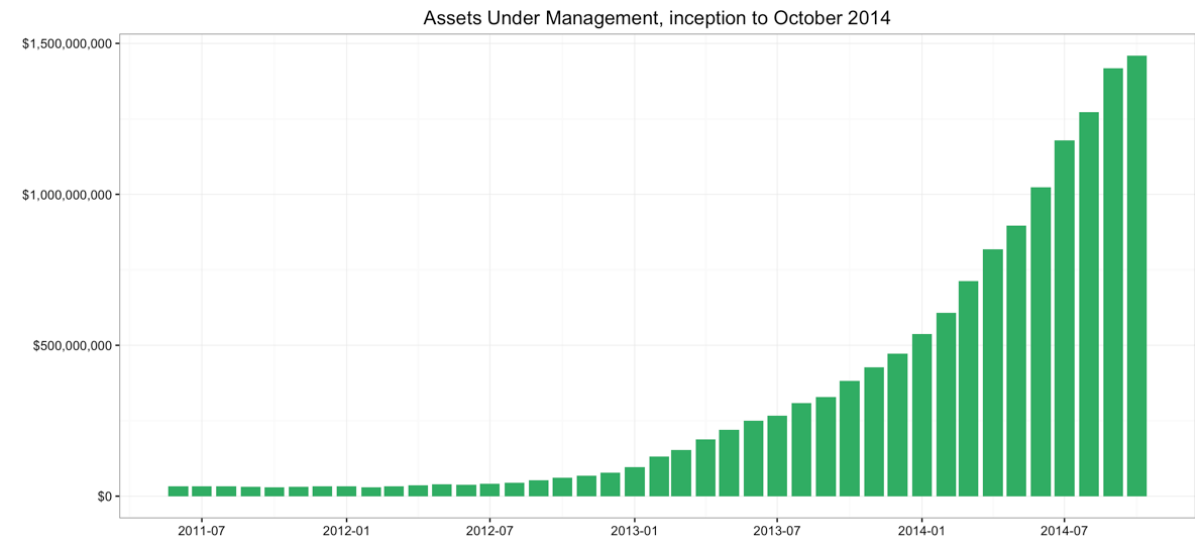
Note: Includes only those who were present as of October 2014.

Exhibit 3 Wealthfront Values



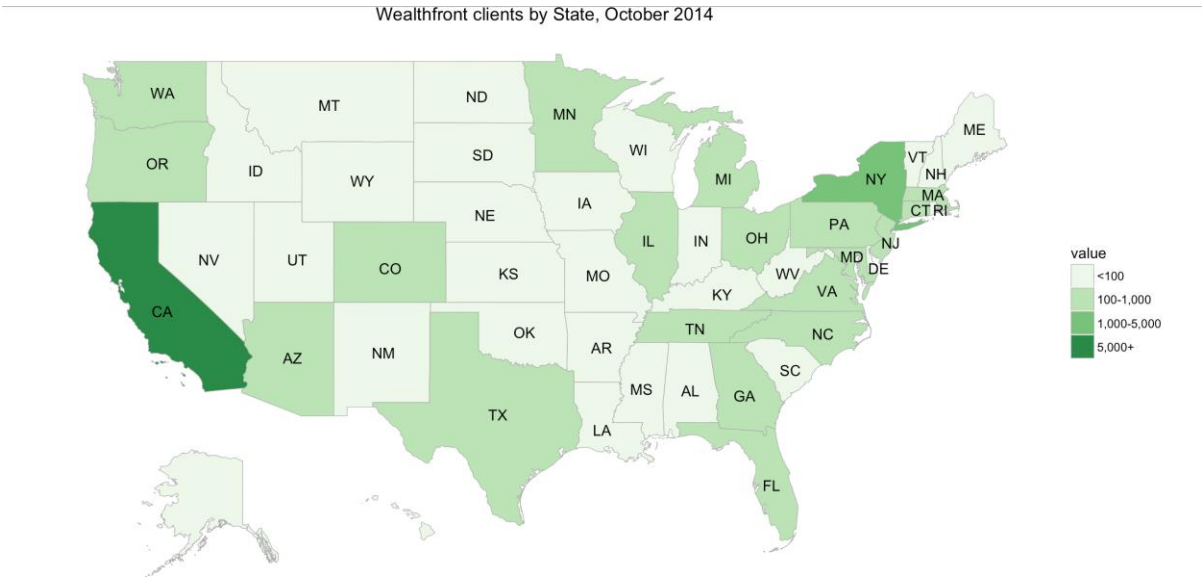
Source: Company website, <https://www.wealthfront.com/investors>, accessed May 2016.

Exhibit 4 AUM Chart

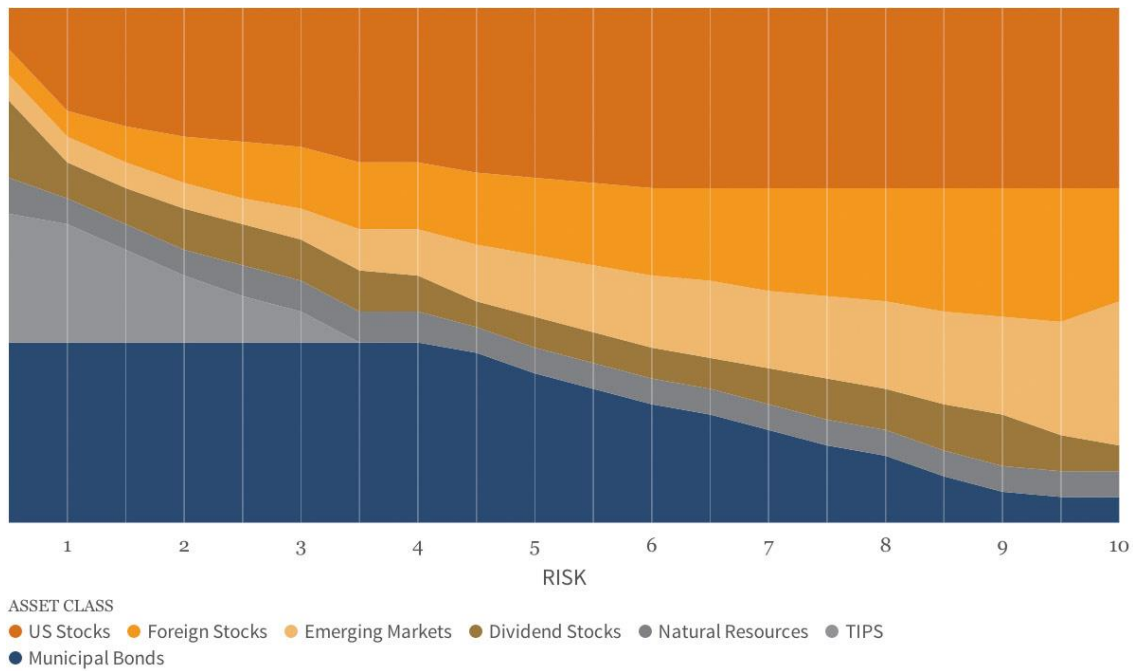


Source: Company documents.

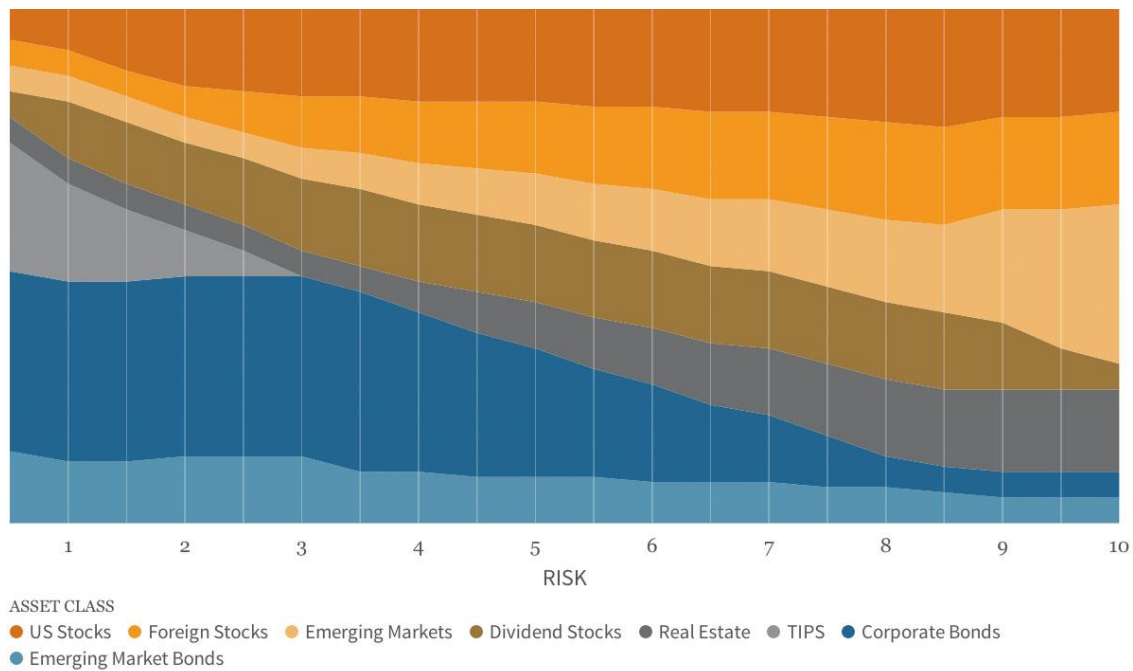
Exhibit 5 Map of U.S. with Client Location Graphic



Source: Company documents.

Exhibit 6a Asset Class Weights by Investor Risk Level for Taxable Accounts

Source: "Wealthfront Investment Methodology White Paper," <https://research.wealthfront.com/whitepapers/investment-methodology/>, accessed May 2016.

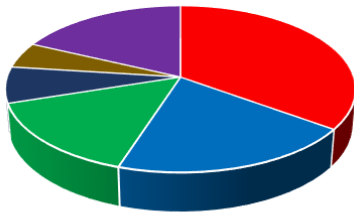
Exhibit 6b Asset Class Weights by Investor Risk Level for Retirement Accounts

Source: "Wealthfront Investment Methodology White Paper," <https://research.wealthfront.com/whitepapers/investment-methodology/>, accessed May 2016.

Exhibit 7 List of ETFs on Wealthfront Platform and Fees

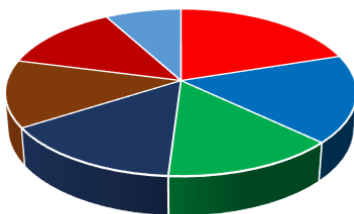
<u>Asset Class</u>	<u>ETF</u>	<u>Ticker</u>	<u>Underlying Index</u>	<u>Expense Ratio</u>
U.S. Stocks	Vanguard	VTI	CRSP US Total Market	0.05%
Foreign Stocks	Vanguard	VEA	FTSE Developed Markets ex. N. America	0.09%
Emerging Markets	Vanguard	VWO	FTSE Emerging Markets	0.15%
Dividend Stocks	Vanguard	VIG	NASDAQ US Dividend Achievers Select / Vanguard Dividend Appreciation	0.10%
Corporate Bonds	iShares	LQD	iBoxx Liquid Investment Grade Corp. Bonds	0.15%
Emerging Market Bonds	iShares	EMB	JP Morgan EMBI Global Core Index	0.40%
Municipal Bonds	iShares	MUB	S&P National Municipal	0.25%
TIPS	Schwab	SCHP	Barclays Capital US TIPS	0.07%
Real Estate	Vanguard	VNQ	Vanguard REIT Index Fund	0.10%
Natural Resources	State Street	XLE	Energy Select Sector	0.14%

Source: Company website, <https://www.wealthfront.com/investors>, accessed May 2016.

Exhibit 8a Sample Model Portfolio with Associated ETFs and Percentage Weightings for Taxable Accounts**Risk Level 7 Sample Portfolio
for Taxable Accounts**

<u>Asset Class</u>	<u>Investment</u>	<u>Percentage</u>	<u>Amount</u>
U.S. Stocks	Vanguard VTI ETF	35%	\$35,000
Foreign Stocks	Vanguard VEA ETF	20%	\$20,000
Emerging Markets	Vanguard VWO ETF	15%	\$15,000
Dividend Stocks	Vanguard VIG ETF	7%	\$7,000
Natural Resources	State Street XLE ETF	5%	\$5,000
Municipal Bonds	iShares MUB ETF	18%	\$18,000

Source: Company documents.

Exhibit 8b Sample Model Portfolio with Associated ETFs and Percentage Weightings for Retirement Accounts**Risk Level 7 Sample Portfolio
for Retirement Accounts**

<u>Asset Class</u>	<u>Investment</u>	<u>Percentage</u>	<u>Amount</u>
U.S. Stocks	Vanguard VTI ETF	20%	\$20,000
Foreign Stocks	Vanguard VEA ETF	17%	\$17,000
Emerging Markets	Vanguard VWO ETF	14%	\$14,000
Dividend Stocks	Vanguard VIG ETF	15%	\$15,000
Real Estate	Vanguard VNQ ETF	13%	\$13,000
Corporate Bonds	iShares LQD ETF	13%	\$13,000
Emerging Market Bonds	iShares EMB ETF	8%	\$8,000
Bonds			

Source: Company documents.

Exhibit 9a Tax-Loss Harvesting Strategy Framework

$$\text{Benefit} - \text{Cost} \geq \text{Threshold}$$

Definitions:

Benefit is calculated by multiplying the potential realized capital loss incurred from selling an ETF times either the short-term or long-term capital gains tax rate, depending on the ETF's holding period.

Cost captures the trading cost of selling the primary ETF and buying the secondary ETF. (Note: this cost is merely the bid-ask spread of the two ETFs because Wealthfront clients never incur any commissions.)

Threshold is an estimate of the expected future harvesting benefit which is modeled by assessing the likelihood that a better harvesting opportunity is available by waiting longer to perform the tax-loss harvesting trade (and thus potentially capturing more of the ETF's decline). Expected return and volatility of each asset class are used as parameters in this estimate.

Source: "Wealthfront Investment Methodology White Paper," <https://research.wealthfront.com/whitepapers/investment-methodology/>, accessed May 2016.

Exhibit 9b Tax Alpha Calculation Methodology

$$\text{Tax Alpha} = \frac{((\text{STCL} \cdot \text{STTR}) + (\text{LTCL} \cdot \text{LTTR}))}{\text{Portfolio Beginning Balance}}$$

Definitions:

STCL is the short-term net capital loss realized.

STTR is the combined federal and state short-term capital gains tax rate.

LTCL is the long-term net capital loss realized.

LTTR is the combined federal and state long-term capital gains tax rate.

Portfolio Beginning Balance is the value of the portfolio at the beginning of each year.

Source: "Wealthfront Investment Methodology White Paper," <https://research.wealthfront.com/whitepapers/investment-methodology/>, accessed May 2016.

Exhibit 10 Innovation Timeline

Source: Andy Rachleff, "Software Based Companies Should be Judged Based on Their Rate of Innovation," <https://blog.wealthfront.com/software-based-companies-judged-rate-innovation/>, accessed May 2016.

Note: The bars illustrate the number of months Wealthfront's innovations preceded similar innovations at other automated investment advisory firms.

Exhibit 11 Competitive Landscape Chart with Brief Company Descriptions

<u>Name</u>	<u>Headquarters</u>	<u>Inception</u>	<u>Account Min.</u>	<u>Fees</u>	<u>Level of Automation</u>
Betterment	New York, NY	2010	None	0.15% to 0.35% depending on account size.	Clients with over \$500,000 eligible for one-time personal consultation.
Charles Schwab Intelligent Portfolio	San Francisco, CA	2015	\$5,000	Free; fixed cash allocation in asset allocation model.	Fully automated.
Future Advisor	San Francisco, CA	2010	\$10,000	Free portfolio analysis and asset allocation recommendation. 0.5% fee for premium services.	Portions fully automated; access to a human financial planner available.
Personal Capital	Redwood City, CA	2011	None for recommendations. \$100,000 for managed portfolios.	Investment recommendations are free of charge. Managed portfolios charged 0.49% to 0.89% depending on account size.	Offers human advisor in addition to automated platform.
SigFig	San Francisco, CA	2012	\$2,000	Portfolio tracking free of charge. Managed portfolios charged 0.25% on accounts over \$10,000.	Offers email contact, phone, and live chat.
Vanguard Personal Advisor Services	Malvern, PA	2013	\$50,000	0.30%	Offers human advisor in addition to automated platform.
Wealthfront	Palo Alto, CA	2011	\$5,000	0.25% on accounts greater than \$25,000.	Fully automated.
Wise Banyan	New York, NY	2013	None	Portfolio advice is free of charge; clients opt in to paid services.	Fully automated.

Source: Company websites.

Endnotes

¹ “Wealthfront,” Case E477, Stanford Business School, May 13, 2013.

² Samantha Raphaelson, “Amid the Stereotypes, Some Facts about Millennials,” NPR New Boom Series, November 18, 2014, <http://www.npr.org/2014/11/18/354196302/amid-the-stereotypes-some-facts-about-millennials>, accessed December 1, 2014.

³ Wealthfront management.

⁴ “Mobile Millennials,” Nielsen Report, September 5, 2014, <http://www.nielsen.com/us/en/insights/news/2014/mobile-millennials-over-85-percent-of-generation-y-owns-smartphones.html>, accessed October 15, 2014.

⁵ Council of Economic Advisors, “The Millennial Report,” October, 2014, https://www.whitehouse.gov/sites/default/files/docs/millennials_report.pdf, accessed November 10, 2014.

⁶ Blake Ellis, “Millennials Love Cash,” Money Section, CNN.com, July 21, 2014.

⁷ Patrick May, “Investing for a New Generation: Mobillennials,” *San Jose Mercury News*, January 11, 2015.

⁸ Wealthfront (blog).

⁹ Wealthfront, “Not Everyone Wants to Manage Their Own Investments” (blog post), July 29, 2015.

¹⁰ “Millennials as Brand Advocates,” SocialChorus, Inc., 2013, <http://info.socialchorus.com/rs/socialchorus/images/Millennials-as-Brand-Advocates-Ebook-SocialChorus.pdf>.

¹¹ Elizabeth MacBride, “Burton Malkiel Joins Wealthfront as CIO,” Wealthfront Knowledge Center (blog post), November 12, 2012, <https://blog.wealthfront.com/burton-malkiel-joins-wealthfront-as-cio/>.

¹² Adam Bryant, “Teaching All Employees to Keep Score,” Interview with Adam Nash, *New York Times*, June 14, 2014.

¹³ Wealthfront Investment Methodology White Paper.

¹⁴ Wealthfront Investment Methodology White Paper.

¹⁵ Jesse Solomon, “Battle Royale: Active Vs. Passive Investing,” CNN Money, December 7, 2014.

¹⁶ Robert D. Arnott, Andrew L. Berkin, and Jia Ye, “How Well Have Taxable Investors Been Served in the 1980’s and 1990’s?,” *Journal of Investment Management* 26 (Summer 2000): 7.

¹⁷ Investment Company Institute Resources on ETF’s website, http://www.ici.org/etf_resources/background/faqs_etfs_market, accessed March 1, 2015.

¹⁸ Wealthfront, “Software Companies Should be Judged Based on their Rate of Innovation,” blog.

¹⁹ Halah Touryalai, “The Biggest, Fastest Growing, and Emergent Talent,” *Forbes*, April 10, 2013.